

Salary Prediction Model and Compensation Fairness Analysis

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05/24/2026

Executive Summary

This analysis examined employee hierarchy, compensation structure, and potential workplace bias within the company using organizational and employee-level data. The primary objectives were to classify employees by organizational level, measure management responsibility, identify the primary factors influencing salary, and evaluate whether compensation practices appeared fair across employee groups.

Employees were successfully classified into six organizational levels: Individual Contributors (IC), Middle Managers (MM), Directors (D), Vice Presidents (VP), Executives (E), and CEO. The resulting hierarchy structure reflected a realistic corporate organization, with the majority of employees classified as Individual Contributors and progressively fewer employees occupying higher leadership positions.

Compensation analysis revealed that salary generally increased with organizational responsibility. Average salaries rose consistently from approximately \$188,000 for Individual Contributors to \$700,000 for the CEO position, indicating that leadership level is a major determinant of employee compensation.

A linear regression salary prediction model was developed using years of experience, department, degree level, signing bonus status, sex, and organizational level as predictors. The model achieved an $R^2 = 0.355$, indicating that approximately 35.5% of salary variation could be explained by the available variables. Organizational level and years of experience emerged as the strongest drivers of salary.

Initial descriptive statistics showed some salary differences between male and female employees at lower organizational levels. However, after controlling for experience, department, organizational level, education, and signing bonus status through regression analysis, the impact of sex on salary became relatively small and not practically significant.

Overall, the findings suggest that compensation within the company is primarily driven by legitimate business factors such as leadership responsibility and experience rather than systematic gender-based discrimination. Nevertheless, continued compensation monitoring and periodic fairness audits are recommended to ensure long-term pay equity and transparency across the organization.

Salary Prediction Model

A linear regression model was developed to predict employee salary using the following variables:

- Years of experience
- Signing bonus status
- Sex
- Degree level
- Department
- Organizational level

The purpose of the model was to identify the primary factors influencing employee compensation and determine whether compensation differences existed after controlling for legitimate business factors.

The model achieved an $R^2 = 0.355$, indicating that approximately 35.5% of salary variation could be explained by the variables included in the analysis.

While the model does not explain all salary variation, this result is reasonable for real-world compensation data, where additional factors such as employee performance, tenure, negotiation outcomes, specialized skills, and market conditions also influence salary decisions.

Key Salary Drivers

The regression analysis identified organizational level and years of experience as the strongest predictors of employee salary.

Average salary increased consistently across organizational levels:

Level Average Salary

IC \$187,909

MM \$192,474

D \$212,344

VP \$258,029

E \$562,500

CEO \$700,000

This suggests that compensation is strongly associated with organizational responsibility and leadership level.

Additionally, employees with greater years of experience generally earned higher salaries, further supporting the conclusion that compensation is primarily driven by legitimate business-related factors.

Gender Compensation Analysis

Salary comparisons by organizational level initially showed some differences between male and female employees, particularly at the Individual Contributor and Middle Manager levels.

However, regression analysis was used to control for other important compensation factors, including:

- Organizational level
- Department
- Years of experience
- Degree level
- Signing bonus status

After controlling for these variables, the coefficient associated with sex was relatively small compared to overall salary levels. The model estimated that male employees earned approximately \$1,863 less than female employees on average when holding all other variables constant.

Given that average salaries in the company ranged from approximately \$180,000 to over \$250,000 for most employee levels, this difference is not practically significant.

Fairness Assessment

Based on the analysis, there is limited evidence of systematic compensation discrimination based on sex within the company.

The results suggest that:

- Organizational level is the primary driver of compensation.
- Years of experience also significantly impact salary.
- Salary differences between male and female employees are substantially reduced after controlling for legitimate business factors.

While some descriptive salary differences were observed at lower organizational levels, the regression analysis indicates that these differences are largely explained by experience, department, and organizational structure rather than gender alone.

Conclusion

This analysis successfully evaluated the company's organizational structure, compensation patterns, and potential indicators of workplace bias using employee hierarchy and salary data.

The findings demonstrate that employee compensation is primarily driven by legitimate business-related factors, particularly organizational level and years of experience. Salary progression across the hierarchy was consistent and aligned with increasing leadership responsibility, suggesting that the company maintains a structured compensation framework.

Although initial descriptive statistics showed some salary differences between male and female employees at lower organizational levels, regression analysis revealed that these differences became minimal after controlling for experience, department, education, signing bonus status, and organizational level. The analysis therefore found limited evidence of systematic gender-based compensation discrimination within the company.

At the same time, the study highlights the importance of continued monitoring and transparency in compensation practices. Expanding future analyses to include additional variables such as employee performance, tenure, promotion history, and geographic pay differences would provide an even more comprehensive understanding of compensation equity.

Overall, the results suggest that the company's compensation structure is largely aligned with organizational responsibility and employee qualifications, while also emphasizing the importance of maintaining ongoing diversity, equity, and inclusion initiatives to support long-term workplace fairness.

Recommendations

The following recommendations are provided to the Head of HR:

1. Continue monitoring compensation equity through periodic data-driven salary audits.
2. Investigate lower-level compensation differences further to ensure consistent pay practices.
3. Improve data collection by incorporating additional variables such as performance ratings, tenure, promotion history, and geographic pay adjustments.
4. Maintain transparent compensation frameworks tied to organizational responsibility and experience.
5. Continue promoting diversity within leadership positions to ensure equitable career advancement opportunities across all employee groups.